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TREND-BASED MOCK PAPER

GATE Mechanical Engineering

Mock Paper — 03

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Mechanical Engineering
Paper:	Mock 03
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-MCQ] [1 mark]

Area of circle radius r :

- (A) $2\pi r$
- (B) πr^2
- (C) πr
- (D) $4\pi r^2$

Q2. [1-MCQ] [1 mark]

Antonym of 'prolific':

- (A) Scarce
- (B) Abundant
- (C) Creative
- (D) Unusual

Section B: Mechanical Engineering

Q3. [1-MCQ] [1 mark]

Derivative of $\sin(x)$:

- (A) $\cos(x)$
- (B) $-\sin(x)$
- (C) $-\cos(x)$
- (D) $\tan(x)$

Q4. [1-MCQ] [1 mark]

The vector $(1, 2, 3)$ projected onto $(2, 1, 0)$ is:

- (A) $(4/5, 2/5, 0)$
- (B) $(8/5, 4/5, 0)$
- (C) $(2, 1, 0)$

(D) (1,2,3)

Q5. [2-NAT] [2 marks]

The area of a circle of radius r is:

(Numerical Answer Type — Enter the numerical value)

Q6. [2-MCQ] [2 marks]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

- (A) $y = 0$
- (B) $y = x \ln(x)$
- (C) $y = x \sin(\ln x)$
- (D) $y = x^2$

Q7. [2-NAT] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

(Numerical Answer Type — Enter the numerical value)

Q8. [2-NAT] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

(Numerical Answer Type — Enter the numerical value)

Q9. [2-MCQ] [2 marks]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q10. [2-NAT] [2 marks]

The rank of the matrix $\begin{bmatrix} 1,2,3 \\ 2,4,6 \\ 3,6,9 \end{bmatrix}$ is:

(Numerical Answer Type — Enter the numerical value)

Q11. [2-MCQ] [2 marks]

$d/dx [\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
- (B) 2
- (C) 4
- (D) 1

Q12. [2-MCQ] [2 marks]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

- (A) 0.68
- (B) 0.32
- (C) 0.95
- (D) 0.16

Q13. [2-MSQ] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

- (A) -1
- (B) -2

- (C) 1
- (D) 2

Q14. [2-NAT] [2 marks]

Which of the following is a probability generating function for Poisson(λ)?

(Numerical Answer Type — Enter the numerical value)

Q15. [2-NAT] [2 marks]

Rolle's theorem requires $f(x)$ to be:

(Numerical Answer Type — Enter the numerical value)

Q16. [1-MCQ] [1 mark]

Laminar flow has:

- (A) Low Re
- (B) High Re
- (C) $Re=1$
- (D) $Re=0$

Q17. [1-NAT] [1 mark]

Bernoulli applies to:

(Numerical Answer Type — Enter the numerical value)

Q18. [1-NAT] [1 mark]

Bernoulli total head =

(Numerical Answer Type — Enter the numerical value)

Q19. [2-NAT] [2 marks]

Boundary layer at $x=1\text{m}$, $Re_x=5e5$ (Blasius):

(Numerical Answer Type — Enter the numerical value)

Q20. [2-NAT] [2 marks]

Laminar pipe flow: Hagen-Poiseuille h_f =

(Numerical Answer Type — Enter the numerical value)

Q21. [2-NAT] [2 marks]

Dynamic viscosity of water at 20°C approx:

(Numerical Answer Type — Enter the numerical value)

Q22. [2-NAT] [2 marks]

Water 5cm pipe at 3 m/s. Re ($\nu=1e-6$):

(Numerical Answer Type — Enter the numerical value)

Q23. [2-NAT] [2 marks]

Pitot tube measures:

(Numerical Answer Type — Enter the numerical value)

Q24. [1-MCQ] [1 mark]

First law based on:

- (A) Energy conservation

- (B) Entropy
- (C) Temperature
- (D) Pressure

Q25. [1-MCQ] [1 mark]

Entropy change for reversible process:

- (A) Positive
- (B) Negative
- (C) Zero
- (D) Infinity

Q26. [1-NAT] [1 mark]

Isothermal expansion: $V_2=2V_1$, $T=300\text{K}$. Work per mole:

(Numerical Answer Type — Enter the numerical value)

Q27. [2-MCQ] [2 marks]

Max efficiency of Carnot between 500K and 300K:

- (A) 40%
- (B) 50%
- (C) 60%
- (D) 80%

Q28. [2-MCQ] [2 marks]

$C_p - C_v$ for ideal gas =

- (A) 0
- (B) R
- (C) 2R
- (D) R/2

Q29. [2-NAT] [2 marks]

For Otto cycle, efficiency depends on:

(Numerical Answer Type — Enter the numerical value)

Q30. [2-MCQ] [2 marks]

Entropy change of universe for reversible process:

- (A) >0
- (B) $=0$
- (C) <0
- (D) Depends

Q31. [2-MSQ] [2 marks]

Carnot engine between 600K and 300K. $Q_{in}=1000\text{J}$. Work output:

- (A) 500J
- (B) 333J
- (C) 667J
- (D) 1000J

Q32. [1-MCQ] [1 mark]

Stefan-Boltzmann: $q = ?$

- (A) $<5\sigma$
- (B) $<5\epsilon$

- (C) $\propto T^0$
(D) $\propto T^1$

Q33. [1-MCQ] [1 mark]

Fourier's law $q =$:

- (A) $-k \, dT/dx$
(B) $k \, dT/dx$
(C) $k \, T$
(D) $-k \, T$

Q34. [1-MCQ] [1 mark]

$Bi \ll 1$ implies:

- (A) Fin
(B) Lumped capacitance
(C) S-B law
(D) Newton cooling

Q35. [2-NAT] [2 marks]

Black body at 1000K emissive power:

(Numerical Answer Type — Enter the numerical value)

Q36. [2-NAT] [2 marks]

Natural convection Nu for vertical plate ($10^4 \leq Gr \cdot Pr \leq 10^9$):

(Numerical Answer Type — Enter the numerical value)

Q37. [2-MCQ] [2 marks]

$k=0.5$, $L=0.2\text{m}$, $A=10\text{m}^2$, $dT=50\text{K}$. Q :

- (A) 1250 W
(B) 625 W
(C) 2500 W
(D) 500 W

Q38. [2-MCQ] [2 marks]

Thermal resistance of plane wall:

- (A) $L/(kA)$
(B) kA/L
(C) kL/A
(D) $A/(kL)$

Q39. [2-MCQ] [2 marks]

Shape factor sphere to large enclosure:

- (A) 1
(B) $4\pi r^2$
(C) πr^2
(D) r

Q40. [1-MCQ] [1 mark]

Cam converts:

- (A) Rotary to rotary
(B) Rotary to reciprocating
(C) Recip to recip

(D) None

Q41. [1-MCQ] [1 mark]

Gear ratio is:

- (A) Teeth ratio
- (B) Speed ratio
- (C) Torque ratio
- (D) Power ratio

Q42. [1-NAT] [1 mark]

Governor maintains:

(Numerical Answer Type — Enter the numerical value)

Q43. [1-NAT] [1 mark]

Gyroscopic couple: $I=2$, $\omega_s=3000\text{rpm}$, $\omega_p=20\text{rad/s}$:

(Numerical Answer Type — Enter the numerical value)

Q44. [2-NAT] [2 marks]

Belt power: $P = (T_1 - T_2) \cdot V$. $V =$

(Numerical Answer Type — Enter the numerical value)

Q45. [2-MCQ] [2 marks]

SHM cam follower. Jerk max at:

- (A) Start
- (B) Middle
- (C) End
- (D) Everywhere zero

Q46. [2-MSQ] [2 marks]

4-bar: $s+l$ vs $p+q$. Equality means:

- (A) Grashof
- (B) Non-Grashof
- (C) Change point
- (D) Toggle

Q47. [2-NAT] [2 marks]

Grashof $s+l < p+q$: fixed shortest =

(Numerical Answer Type — Enter the numerical value)

Q48. [1-NAT] [1 mark]

Pattern volume for 1000 cm^3 casting with 2% shrinkage:

(Numerical Answer Type — Enter the numerical value)

Q49. [1-MCQ] [1 mark]

Tresca theory for:

- (A) Brittle
- (B) Ductile
- (C) Both
- (D) Neither

Q50. [1-MCQ] [1 mark]

Orthogonal cutting shear angle:

- (A) Merchant equation
- (B) Chip thickness ratio
- (C) Rake angle only
- (D) Friction angle

Q51. [1-MCQ] [1 mark]

Taylor tool life $VT^n = C$. Speed doubled, life:

- (A) $1/8$
- (B) $1/4$
- (C) $1/2$
- (D) $1/16$

Q52. [2-MSQ] [2 marks]

ECM (Electrochemical Machining) removes metal by:

- (A) Cutting
- (B) Electrochemical dissolution
- (C) Abrasion
- (D) Melting

Q53. [2-NAT] [2 marks]

Six Sigma defects per million:

(Numerical Answer Type — Enter the numerical value)

Q54. [2-NAT] [2 marks]

Lathe: workpiece rotates against:

(Numerical Answer Type — Enter the numerical value)

Q55. [1-NAT] [1 mark]

Cantilever UDL w over L . Deflection:

(Numerical Answer Type — Enter the numerical value)

Q56. [2-MCQ] [2 marks]

SS beam UDL w N/m, span L . Max BM:

- (A) $wL^2/8$
- (B) $wL^2/12$
- (C) $wL^2/4$
- (D) $wL^2/2$

Q57. [2-NAT] [2 marks]

Max shear stress in solid shaft:

(Numerical Answer Type — Enter the numerical value)

Q58. [1-MCQ] [1 mark]

Stress = Force /:

- (A) Length
- (B) Area
- (C) Volume
- (D) Mass

Q59. [1-NAT] [1 mark]

Section modulus $Z =$:

(Numerical Answer Type — Enter the numerical value)

Q60. [1-MCQ] [1 mark]

MIG welding uses:

- (A) Flux
 - (B) Gas shield
 - (C) Both
 - (D) None
-

Q61. [1-MCQ] [1 mark]

Sand casting uses:

- (A) Permanent mold
 - (B) Sand mold
 - (C) Die
 - (D) Investment
-

Q62. [1-MCQ] [1 mark]

Equilibrium requires:

- (A) $\sum F = 0$
 - (B) $\sum M = 0$
 - (C) Both
 - (D) None
-

Q63. [1-MCQ] [1 mark]

$F = ma$ is:

- (A) Newton 1st
 - (B) 2nd
 - (C) 3rd
 - (D) None
-

Q64. [1-NAT] [1 mark]

Key prevents:

(Numerical Answer Type — Enter the numerical value)

Q65. [1-MCQ] [1 mark]

Shaft designed for:

- (A) Torsion
 - (B) Bending
 - (C) Combined
 - (D) Axial
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1MCQ	B	1
2	1MCQ	A	1
3	1MCQ	A	1
4	1MCQ	B	1
5	1NAT	A	1
6	1MCQ	A	1
7	1NAT	B	1
8	1MCQ	B	1
9	1MCQ	B	1
10	1MCQ	A	1
11	1MCQ	A	1
12	1MCQ	D	1
13	1MCQ	A	1

14	1MCQ	B	1
15	1MCQ	B	1
16	1MCQ	C	1
17	1MCQ	B	1
18	1NAT	C	1
19	1MCQ	C	1
20	1NAT	B	1
21	1NAT	A	1
22	1NAT	B	1
23	1NAT	A	1
24	1NAT	A	1
25	1MCQ	B	1
26	1NAT	C	1
27	1MCQ	B	1
28	1MCQ	A	1

29	1MCQ	A	1
30	1MCQ	B	1
31	2NAT	A	2
32	2NAT	A	2
33	2NAT	B	2
34	2MCQ	A	2
35	2MCQ	A	2
36	2NAT	A	2
37	2NAT	A	2
38	2MCQ	B	2
39	2NAT	B	2
40	2NAT	A	2
41	2MCQ	A	2
42	2NAT	A	2
43	2NAT	A	2

44	2NAT	B	2
45	2MCQ	A	2
46	2MCQ	C	2
47	2MSQ	C	2
48	2MCQ	B	2
49	2NAT	A	2
50	2MSQ	B	2
51	2NAT	B	2
52	2NAT	B	2
53	2MSQ	A	2
54	2NAT	A	2
55	2MCQ	B	2
56	2NAT	C	2
57	2NAT	A	2
58	2MCQ	C	2

59	2MCQ	A	2
60	2MCQ	A	2
61	2MCQ	A	2
62	2MSQ	A	2
63	2NAT	A	2
64	2NAT	A	2
65	2NAT	C	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Quantitative | Difficulty: difficult

Q2.

Answer: A

Solution:

Correct Answer: A

Subject: General Aptitude | Topic: Verbal | Difficulty: easy

Q3.

Answer: A

Solution:

Correct Answer: A

Subject: Engineering Mathematics | Topic: Calculus | Difficulty: easy

Q4.

Answer: B

Solution:

Correct Answer: B

Subject: Strength of Materials | Topic: Stress | Difficulty: moderate

Q5.

Answer: A

Solution:

Correct Answer: A

Subject: Strength of Materials | Topic: Bending | Difficulty: easy

Q6.

Answer: A

Solution:

Correct Answer: A

Subject: Fluid Mechanics | Topic: Viscosity | Difficulty: moderate

Q7.

Answer: B

Solution:

Correct Answer: B

Subject: Fluid Mechanics | Topic: Bernoulli | Difficulty: easy

Q8.

Answer: B

Solution:

Correct Answer: B

Subject: Manufacturing Processes | Topic: Welding | Difficulty: easy

Q9.

Answer: B

Solution:

Correct Answer: B

Subject: Manufacturing Processes | Topic: Casting | Difficulty: moderate

Q10.

Answer: A

Solution:

Correct Answer: A

Subject: Thermodynamics | Topic: First Law | Difficulty: moderate

Q11.

Answer: A

Solution:

Correct Answer: A

Subject: Thermodynamics | Topic: Entropy | Difficulty: easy

Q12.

Answer: D

Solution:

Correct Answer: D

Subject: Heat Transfer | Topic: Radiation | Difficulty: easy

Q13.

Answer: A

Solution:

Correct Answer: A

Subject: Heat Transfer | Topic: Conduction | Difficulty: moderate

Q14.

Answer: B

Solution:

Correct Answer: B

Subject: Theory of Machines | Topic: Cam | Difficulty: moderate

Q15.

Answer: B

Solution:

Correct Answer: B

Subject: Theory of Machines | Topic: Gears | Difficulty: moderate

Q16.

Answer: C

Solution:

Correct Answer: C

Subject: Engineering Mechanics | Topic: Statics | Difficulty: easy

Q17.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mechanics | Topic: Dynamics | Difficulty: easy

Q18.

Answer: C

Solution:

Correct Answer: C

Subject: Machine Design | Topic: Keys | Difficulty: easy

Q19.

Answer: C

Solution:

Correct Answer: C

Subject: Machine Design | Topic: Shaft | Difficulty: moderate

Q20.

Answer: B

Solution:

Cantilever UDL: $\delta = \frac{wL^4}{8EI}$.

Subject: SOM | Topic: SOM | Difficulty: moderate

Q21.

Answer: A

Solution:

$W = nRT \ln(V_2/V_1) = 8.314 \times 300 \times \ln(2) = 2494.2 \times 0.693 = 1729 \text{ J}$.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: moderate

Q22.

Answer: B

Solution:

Total head = $P/\rho + V^2/2 + gz$.

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: moderate

Q23.

Answer: A

Solution:

Governor: automatic speed regulator. Maintains constant engine speed under varying load.

Subject: Theory of Machines | Topic: Theory of Machines | Difficulty: easy

Q24.

Answer: A

Solution:

Pattern = Casting/(1-0.02) = 1000/0.98 = 1020.4 cm³.

Subject: Manufacturing | Topic: Manufacturing | Difficulty: moderate

Q25.

Answer: B

Solution:

proj = ((v.w)/(w.w))*w = (4/5)*(2,1,0) = (8/5, 4/5, 0).

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q26.

Answer: C

Solution:

C = I*omega_s*omega_p = 2*314.16*20 = 12566 Nm.

Subject: Theory of Machines | Topic: Theory of Machines | Difficulty: easy

Q27.

Answer: B

Solution:

Tresca (max shear stress): ductile materials.

Subject: Manufacturing | Topic: Manufacturing | Difficulty: difficult

Q28.

Answer: A

Solution:

$\tan(\phi) = (r \cdot \cos(\alpha)) / (1 - r \cdot \sin(\alpha))$. Merchant's equation.

Subject: Manufacturing | Topic: Manufacturing | Difficulty: easy

Q29.

Answer: A

Solution:

$T_2/T_1 = (V_1/V_2)^{1/n}$. For n=0.5: $(1/2)^2 = 1/4$. For n=0.25: $(1/2)^4 = 1/16$.

Subject: Manufacturing | Topic: Manufacturing | Difficulty: moderate

Q30.

Answer: B

Solution:

Bi < 0.1: lumped capacitance valid.

Subject: Heat Transfer | Topic: Heat Transfer | Difficulty: easy

Q31.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q32.

Answer: A

Solution:

$V = \pi D N / 60$ = linear velocity of belt.

Subject: Theory of Machines | Topic: Theory of Machines | Difficulty: difficult

Q33.

Answer: B

Solution:

$\delta_{99} = 5x/\sqrt{\text{Re}} = 5 \times 1000/\sqrt{5e5} = 7.07\text{mm}$. δ_{99} (99%) approx 7mm. Close to B.

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: difficult

Q34.

Answer: A

Solution:

SS UDL: max BM at center = $wL^2/8$.

Subject: SOM | Topic: SOM | Difficulty: moderate

Q35.

Answer: A

Solution:

$\eta = 1 - 300/500 = 1 - 0.6 = 0.4 = 40\%$.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: easy

Q36.

Answer: A

Solution:

$\tau_{\text{max}} = Tc/J = 16T/(\pi d^3)$.

Subject: SOM | Topic: SOM | Difficulty: moderate

Q37.

Answer: A

Solution:

Laminar: $h_f = 32\mu L^2 V/(\rho g D^4)$.

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: easy

Q38.

Answer: B

Solution:

Mayer's relation: $C_p - C_v = R$ for ideal gas.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: moderate

Q39.

Answer: B

Solution:

Otto: $\eta = 1 - r^{1-\gamma}$. Depends on compression ratio r and γ .

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: moderate

Q40.

Answer: A

Solution:

$E = \sigma T^4 = 5.67e-8 \times 1e12 = 5.67e4 = 56.7 \text{ kW/m}^2$.

Subject: Heat Transfer | Topic: Heat Transfer | Difficulty: easy

Q41.

Answer: A

Solution:

Homogeneous ODE. Substituting $y = x^m$ and $y(1)=0$ gives $m=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q42.

Answer: A

Solution:

Vertical plate laminar: $Nu = 0.59(GrPr)^{0.25}$.

Subject: Heat Transfer | Topic: Heat Transfer | Difficulty: difficult

Q43.

Answer: A

Solution:

$\mu_{\text{water}}(20^\circ\text{C})$ approx $1e-3 \text{ Pa.s}$ or 1 cP .

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: easy

Q44.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q45.

Answer: A

Solution:

$Q = k \cdot A \cdot \Delta T / L = 0.5 \cdot 10 \cdot 50 / 0.2 = 1250 \text{ W}$.

Subject: Heat Transfer | Topic: Heat Transfer | Difficulty: moderate

Q46.

Answer: C

Solution:

SHM: jerk = da/dt . Max at start/end where acceleration changes direction.

Subject: Theory of Machines | Topic: Theory of Machines | Difficulty: easy

Q47.

Answer: C

Solution:

$s+l = p+q$: change point mechanism. All four links can become cranks.

Subject: Theory of Machines | Topic: Theory of Machines | Difficulty: moderate

Q48.

Answer: B

Solution:

Reversible: $\Delta S_{\text{univ}} = 0$. Irreversible: $\Delta S_{\text{univ}} > 0$ (second law).

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: easy

Q49.

Answer: A

Solution:

$Re = 3 \cdot 0.05 / 1e-6 = 150000 = 1.5e5$.

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: difficult

Q50.

Answer: B

Solution:

ECM: anodic dissolution. Metal removed by electrochemical reaction, no tool wear.

Subject: Manufacturing | Topic: Manufacturing | Difficulty: moderate

Q51.

Answer: B

Solution:

Grashof + fixed shortest: crank-rocker mechanism.

Subject: Theory of Machines | Topic: Theory of Machines | Difficulty: moderate

Q52.

Answer: B

Solution:

$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q53.

Answer: A

Solution:

$\eta = 1 - 300/600 = 0.5$. $W = 0.5 \cdot 1000 = 500 \text{ J}$.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: easy

Q54.

Answer: A

Solution:

Six Sigma: 3.4 DPMO. 99.99966% yield.

Subject: Manufacturing | Topic: Manufacturing | Difficulty: easy

Q55.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q56.

Answer: C

Solution:

Pitot: stagnation/total pressure.

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: difficult

Q57.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q58.

Answer: C

Solution:

$\frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $\frac{d}{dx} \ln(\sec x) = \tan x$. $\frac{d}{dx} \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = \frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q59.

Answer: A

Solution:

$R_{th} = L/(k*A)$.

Subject: Heat Transfer | Topic: Heat Transfer | Difficulty: easy

Q60.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q61.

Answer: A

Solution:

$F_{12} = 1$ for sphere to infinite enclosure.

Subject: Heat Transfer | Topic: Heat Transfer | Difficulty: moderate

Q62.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q63.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q64.

Answer: A

Solution:

Lathe: workpiece rotates, static tool feeds.

Subject: Manufacturing | Topic: Manufacturing | Difficulty: moderate

Q65.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 13 questions, 24 marks
Fluid Mechanics: 8 questions, 13 marks
Thermodynamics: 8 questions, 13 marks
Heat Transfer: 8 questions, 13 marks
Theory of Machines: 8 questions, 12 marks
Manufacturing: 7 questions, 10 marks
SOM: 3 questions, 5 marks
General Aptitude: 2 questions, 2 marks
Strength of Materials: 2 questions, 2 marks
Manufacturing Processes: 2 questions, 2 marks
Engineering Mechanics: 2 questions, 2 marks
Machine Design: 2 questions, 2 marks

Difficulty Distribution:

Easy: 28 | Moderate: 27 | Difficult: 10

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.